Universidad del Desarrollo

Class 8: Machine Learning – Decision Trees

UNIT III: Advanced Analytics and Machine Learning Models

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What is a Decision Tree in ML?

- **Supervised ML Model:** A predictive algorithm that learns from labeled data to map inputs (X_1,X_2,…,X_n) to targets (Y).

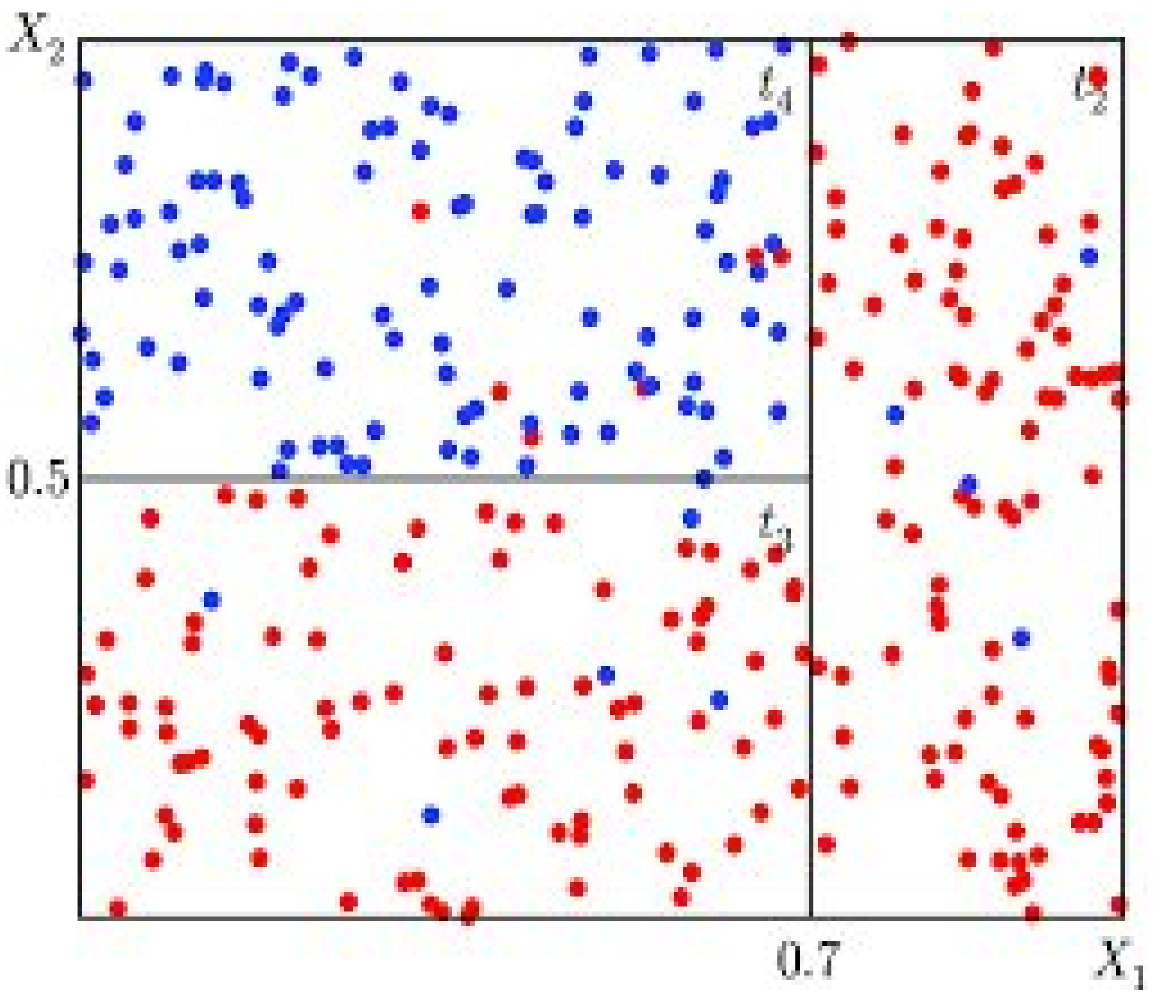
- **Geometry of Logic - Recursive Binary Partitioning:** An iterative process that splits the data into two branches **to maximize homogeneity.**

- **Non-Parametric Model:** No fixed mathematical formula; the model adapts its structure to the data's "shape."

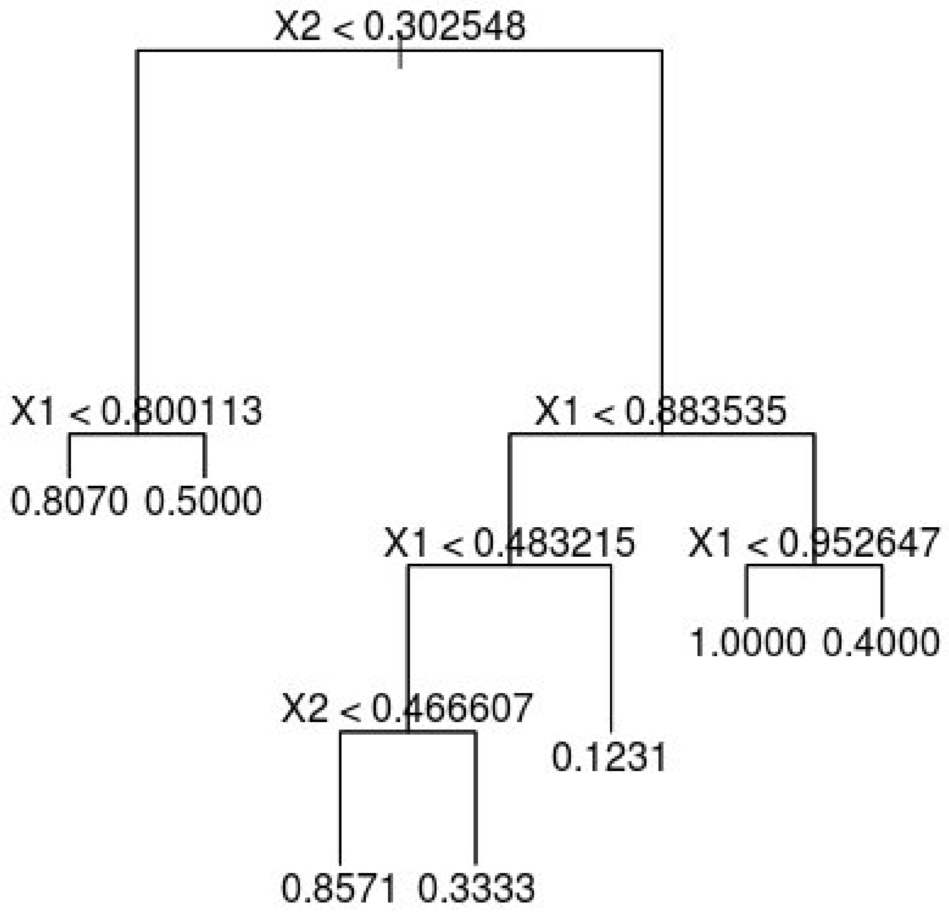
- **Space Segmentation ($\mathbb{R}^n$): 2D** – Rectangles, **3D** - Rectangular Prisms, **nD:** Hyper-rectangles.

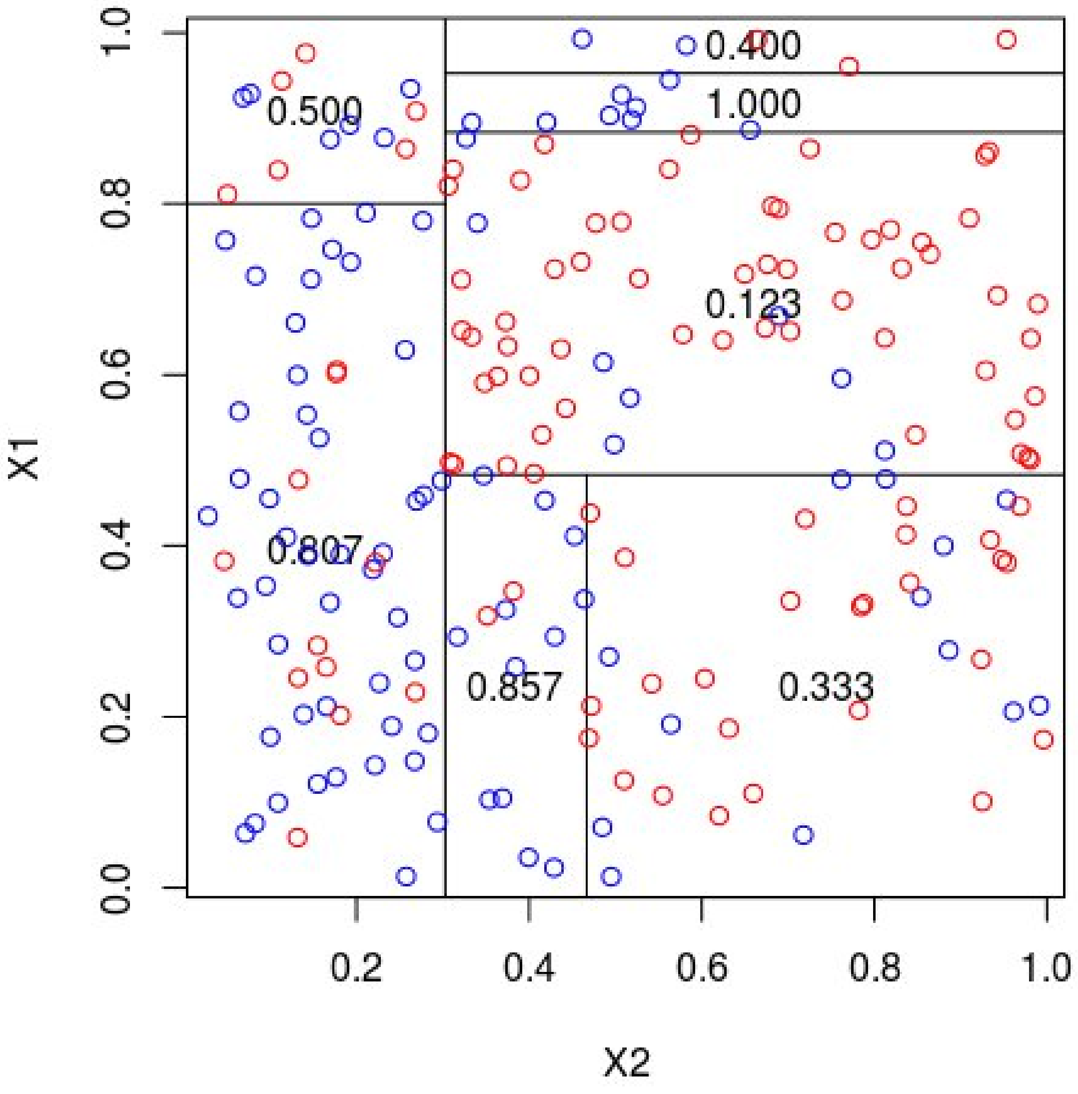
- **The tree is a translator:** It turns an invisible high-dimensional 'hyper-cloud' of points into a transparent flowchart of logical rules (IF/ELSE).

- Useful for both **Classification** (predicting labels) and **Regression** (predicting continuous values).

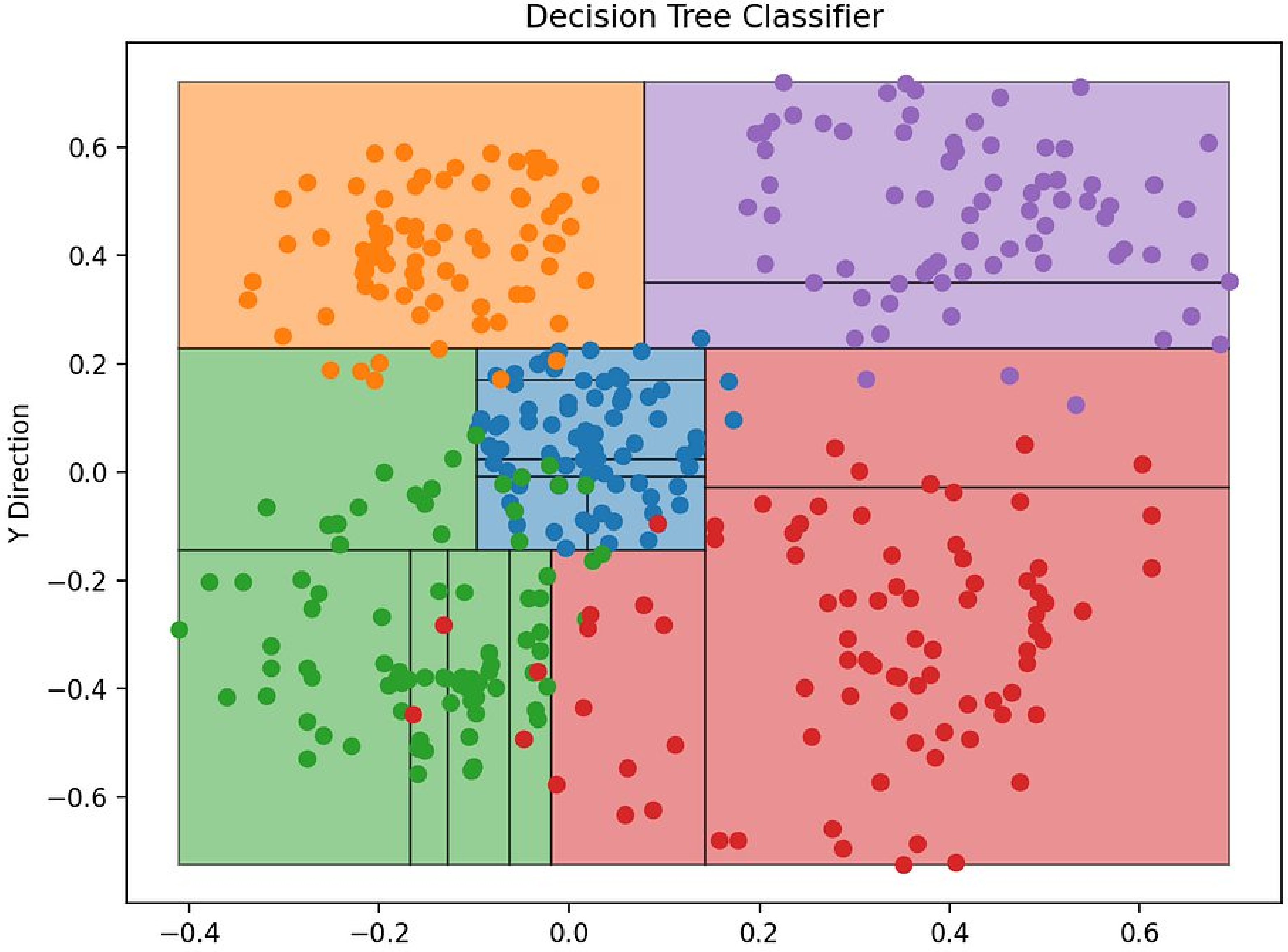


What is a Decision Tree in ML ? Regression Problem





What is a Decision Tree in ML? Classification Problem



Main algorithm structure - Tree Anatomy

- **Root Node (Input):** The entry point of the algorithm. It contains the entire dataset before any partitions are applied.

- **Optimal Split Selection:** At every node, the model performs an exhaustive search to find the **best parameters** $\theta^* = (j,s)$.

Feature (j): The index of the variable X_j that best separates the classes or values.
Threshold (s): The exact point of bisection within the range of X_j.

- **The Partition Rule:** The space is divided into two disjoint subsets (child nodes):

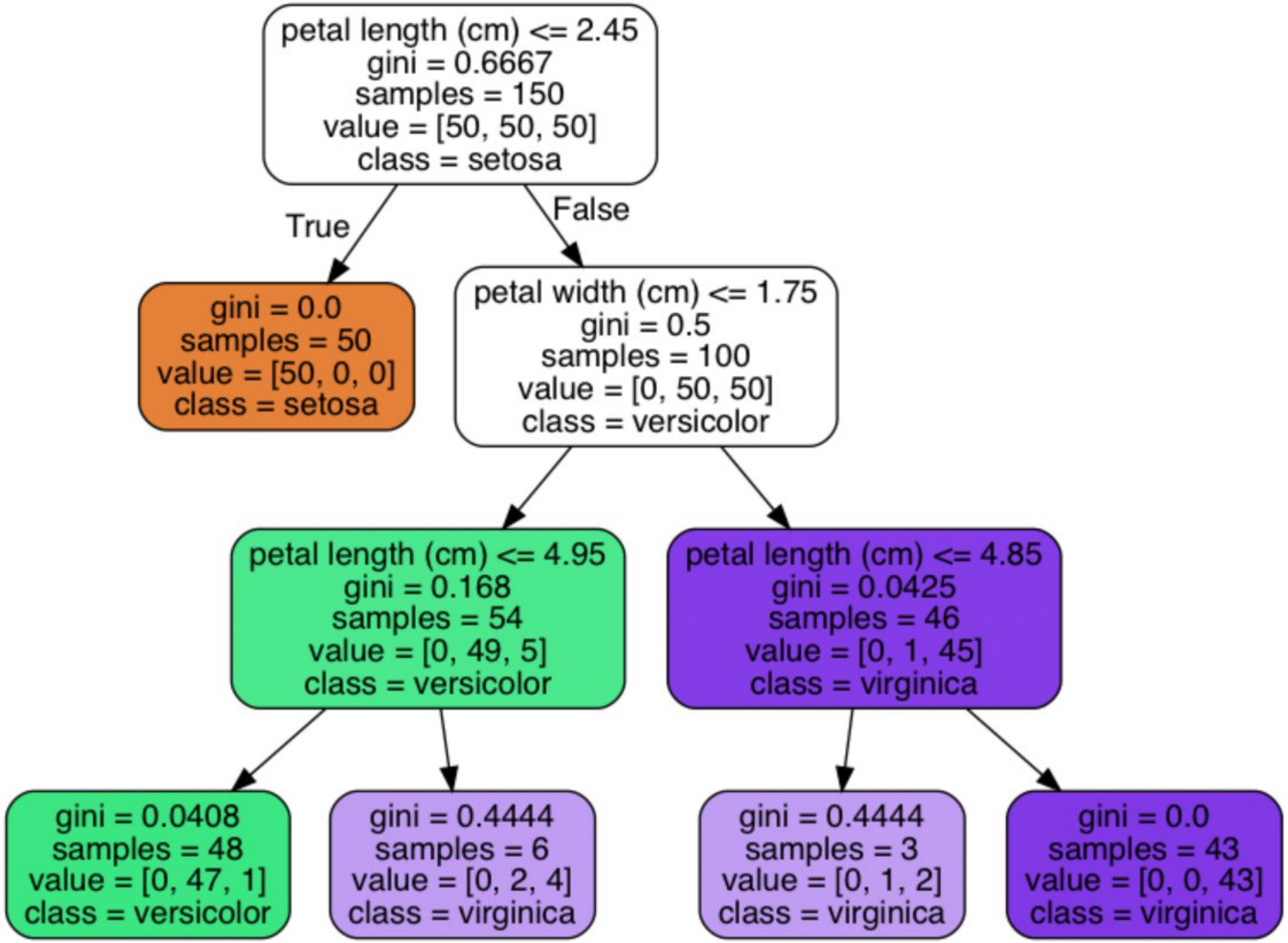
$L(j,s) =$ All samples where $X_j \leq s$ $\qquad$ $R(j,s) =$ All samples where $X_j > s$.

- **Internal Nodes:** Intermediate checkpoints that store the local (j,s) decision logic. They represent the "nested" nature of the model.

- **Leaf nodes:** Terminal regions R_m where the search stops.

Prediction (Regression): $mode(y \in R_m)$ **Prediction (Classification):** $mean(y \in R_m)$.

Main algorithm structure - Tree Anatomy



Optimization in Classification – Slicing process

- At each node, the algorithm identifies the optimal pair $\theta^* = (j,s)$ that minimizes the combined impurity of the resulting branches.

- **Impurity Metric (Gini Index):** For any set of samples A, the impurity is defined as:

$$G(A) = 1 - \sum_{k=0}^{n} p_k^2$$

Where p_k is the proportion of class k within set A.

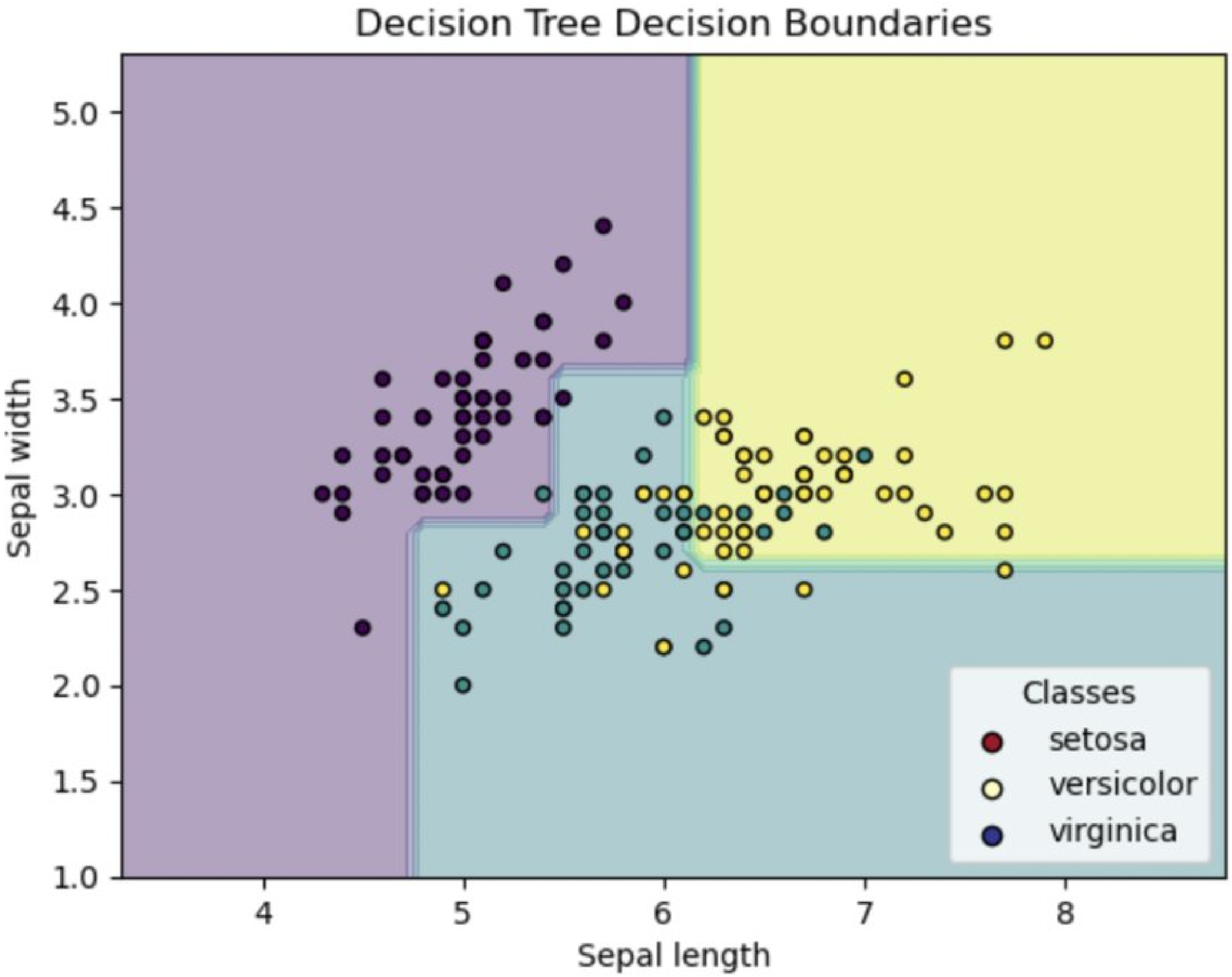
- **Loss function:**

$$Min\left\{C(s,j) = \frac{n_L}{n} G(L(s,j)) + \frac{n_R}{n} G(R(s,j))\right\}$$

- Where n_L and n_L are the number of samples in the left and right children, n is the total in the current node. $L(s,j)$ and $R(s,j)$ are the child nodes (splited regions). The algorithm performs an exhaustive discrete search over all features and all possible split points. It selects the (j,s) coordinates that maximize the 'clumping' of identical labels in the resulting regions.

- The objective is to find the 'cut' that yields the lowest average impurity. We weight the children by their size because we want to ensure that our decisions remain statistically significant for as many samples as possible.

Optimization in Classification – Slicing process



Optimization in Regression – Slicing process

- When predicting numerical values, the goal shifts from "Purity" to **"Variance Reduction."** We seek the split that minimizes the total error within the regions.

- **Error metric (MSE):** For any set of samples A, the "impurity" is defined by the **Mean Squared Error (MSE)**:

$$MSE(A) = \frac{1}{n_A}\sum_{i \in A} (y_i - \bar{y}_A)^2$$

Where $\bar{y}_A$ is the mean of the target values y within set A.

- **Loss function:** The algorithm solves the following optimization problem by iterating over all features j and thresholds s:

$$Min\left\{C(s,j) = \frac{n_L}{n} MSE(L(s,j)) + \frac{n_R}{n} MSE(R(s,j))\right\}$$

- In Regression, the tree operates as a **local step-function generator**. By minimizing MSE, the algorithm carves the feature space into n-dimensional **hyper-rectangles**, assigning the local mean as a constant prediction for each. This allows the model to approximate complex non-linear patterns by breaking them down into a sequence of discrete steps, making it exceptionally robust for capturing sudden shifts in data.

Optimization in Regression – Slicing process

Alternative Impurity Criteria

- The **C**lassification **A**nd **R**egression **T**rees (CART) Framework & Information Gain:

Metric	Domain	Logic for the Model
Gini Impurity	Classification	Measures the probability of a random element being misclassified. **Faster** (no logs).
Entropy	Classification	Measures the "Information Bits" (uncertainty). **Strict**; penalizes mixed nodes more aggressively.
Information Gain	Classification	The **delta** or "profit" in information. The algorithm selects the split that **maximizes** this gain.
MSE / Variance	Regression	Measures the "spread" of continuous values. The goal is to minimize internal variance within the leaf.

Stopping Criteria & Hyperparameters

- A tree with no constraints will grow until every leaf is pure (G=0), leading to **overfitting**. We use hyperparameters to restrict growth and improve generalization.

Hyperparameter	Functional Objective	Operational Impact	Typical Range
max_depth	Limits tree height	Constrains feature interaction depth.	[3,20]
min_samples_split	Pre-split requirement	Prevents splitting small, noisy clusters.	[2,50]
min_samples_leaf	Minimum leaf size	Smooths predictions; reduces variance.	[1,20]
max_leaf_nodes	Caps total complexity	Limits the number of "step-functions."	[10,100+]
criterion	Selects impurity metric	Switches between Gini or Entropy/MSE.	gini, entropy

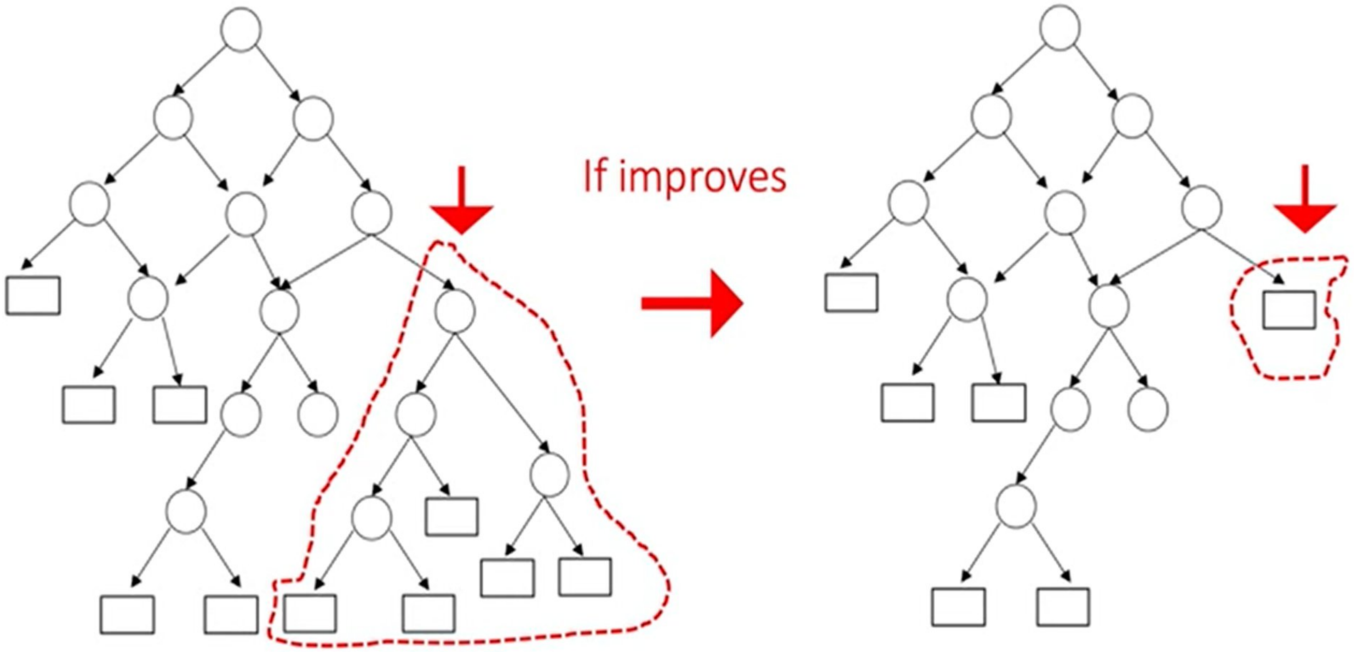
Practical Trade-offs

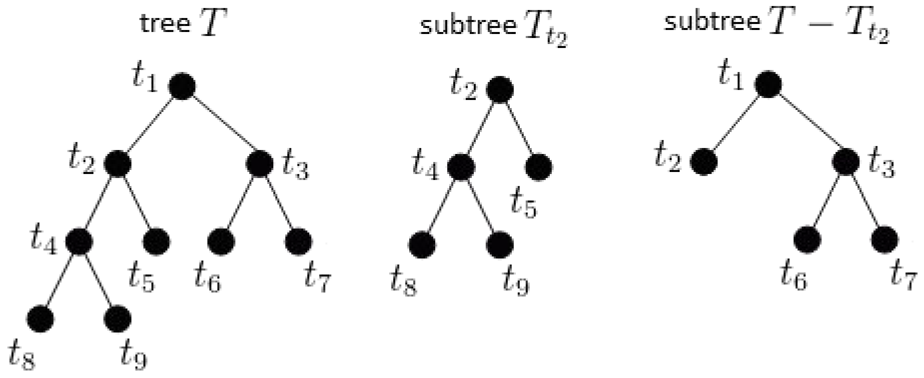
- Decision Trees are "White Box" models, offering high interpretability but facing specific structural limitations that every Data Scientist must manage.
- Decision trees are inherently **unstable and prone to high variance**, meaning minor data fluctuations can lead to completely different splits and poor generalization.

Aspect	Advantage	Constraint
Model Nature	**White-Box:** Decisions are traceable and human-readable.	**High Variance:** Highly sensitive to small training set fluctuations.
Data Topology	**Non-Linear:** Captures complex feature interactions natively.	**Orthogonal Splits:** Can only cut the space in axis-aligned steps.
Preprocessing	**Zero-Effort:** No scaling, centering, or normalization required.	**Greedy Heuristic:** Local optima at each node; no global optimization.
Utility	**Feature Rank:** Automates identification of key drivers.	**Overfitting:** Naturally tends to memorize noise/outliers.

Pruning & Complexity Control

- Pruning is the process of simplifying a tree by removing branches that add noise rather than useful information.
- **Pre-pruning (Early Stopping):** Stops growth early based on preset limits (e.g., depth or sample count). It is the fastest method, but might miss complex patterns that appear in deeper levels.
- **Post-pruning (Cost Complexity Pruning):** The tree grows fully and is then "trimmed" back. It evaluates if the error reduction of a branch justifies its complexity.
- A branch is **collapsed into a leaf** if its contribution to error reduction is less than the complexity "tax" defined by α $(\Delta Error < \alpha)$; this mathematically ensures that only branches with significant predictive power remain in the model.





Model Interpretability & Business Logic

- **Translating Non-Functional Models for Stakeholders:** Since Decision Trees do not have a functional form (like $y = \beta 0 \; + \beta 1 \; x$), we use the following framework to interpret "feature weight" and decision flow:

Tool	Technical Metric	Stakeholder Utility
IF-THEN Rules	Decision Path (Root to Leaf)	**Transparency:** Provides a clear "audit trail" for every individual prediction.
Feature Importance (MDI)	Mean Decrease in Impurity	**Strategy:** Identifies which variables drive the most value across the entire business logic.
SHAP Values	Local Feature Contribution	**Individualization:** Explains exactly why a specific case deviated from the average baseline.

Practical Section

Google Colab — ML Linear Regression v/s Decision Tree
Simple and Multiple scenarios
No linear relationship between features and target
Performance metrics
Visualization

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